

PVLoad-R1 Bench Card

WIRING THE ARDUINO

Connect ten wires from the Arduino to the 10-pin header on the board. Pin 1 is the square pad.

1	5V	5 V
2	GND	GND
3	D6	K1 drive
4	D7	K2 drive
5	D8	K3 drive
6	D9	CS U2
7	D10	CS U1
8	D11	SDI
9	D12	SDO
10	D13	SCK

On the Uno, 5V and GND are on the power side. D8 to D13 are the six pins nearest the USB socket.

THE THREE BANANA JACKS

- J1**, red. PV+, the cell's positive.
- J2**, red. +24 V from the bench supply.
- J3**, black. Ground. Connect both the supply negative and the cell negative here. Use a stacking plug. Do not force two plugs into one jack.

POWER UP

1. 24 V on
2. 5 V on (plug in the Arduino)
3. PV− → J3
4. PV+ → J1
5. illuminate the cell

POWER DOWN

1. darken the cell
2. PV+ out of J1
3. PV− out of J3
4. Arduino out (5 V off)
5. 24 V off

WHAT ON AND OFF MEAN

Do not unplug anything from the board to turn a supply on or off. The supply leads stay in J2 and J3. The ten header wires stay on the Arduino.

For 24 V, press the output button on the bench supply. If the supply has no output button, put the black lead in J3 first, then push the red lead into J2.

For 5 V, plug or unplug the USB cable at the laptop end. Stop the script first.

THE LIGHT IS YOURS TO SET

The script does not control the illumination and does not know what it is. Set the lamp, run the sweep, change the lamp, run it again. Each run writes its own timestamped CSV. Put the lamp setting in RUN_TAG first: nothing else records it, and two runs at different light are otherwise indistinguishable.

USING THE HANDHELD METER

Put the black lead in the socket marked **COM**. Put the red lead in the socket marked **VΩ**. Never use the **10A** socket. That socket makes the meter a piece of wire and shorts J1 to J3.

Turn the dial to **ohms** (Ω) to read the board's resistance across J1 and J3. Turn it to **DC volts** (V with a straight line over dashes) to probe pins against ground.

CABLES TO THE LAPTOP

- Arduino.** A standard USB printer cable, A to B.
- Bench supply.** Two banana leads to J2 and J3. Set 24.0 V and a current limit near 100 mA.
- The two bench meters.** A Keithley 196 and an Agilent 34401A. Neither has USB. Each has one GPIB socket, the wide connector with two screws. Use a single USB to GPIB adapter, Keysight 82357B. Plug the adapter into the laptop. Run a GPIB cable from the adapter to the first meter. GPIB connectors stack, so run a second GPIB cable from the first meter to the second. Both meters share the one adapter.
- Give each meter its own address before you connect them. Two meters cannot share one. Whatever you choose, it has to match DMM_V_ADDRESS and DMM_I_ADDRESS in the script.

WHICH METER GOES WHERE

- Put the **34401A on volts** and the **196 on amps**. Both are ordinary DMMs on current, so that side is a wash. On volts they are not: at the open-circuit point the board presents 470 kΩ, and a 10 MΩ meter across it reads Voc about 4.5% low.
- The 34401A can be told to raise its input above 10 GΩ, and the script sends that command to whichever meter is the voltmeter. The 196 has no such command, and the 30 V range a 9 V cell lands on is 10 MΩ.
- Tell the script which is which with DMM_V_MODEL and DMM_I_MODEL. It asks each meter its model at startup and stops if they disagree.

CONNECTING THE METERS TO THE CELL

The amps meter goes inside the wire. Current must flow through the meter. Cut the wire from the cell's + terminal to J1 and put the meter in the break. Current does not go into the volts terminal on either meter. On the 196 use **AMPS** and **LO**; on the 34401A use the current terminal, the one with the fuse symbol, and **LO**.

cell +	→	AMPS or I
LO	→	J1

The volts meter goes across the cell. Cut nothing. Two leads touch the cell's terminals. On the 196 use **VOLTS OHMS HI** and **LO**; on the 34401A use **HI** and **LO**.

HI	→	cell +
LO	→	cell −

Clip these to the cell's own leads, not to J1 and J3. On the jacks the reading passes through the amps meter and comes out low by its burden drop, which is tens of millivolts on either of these meters. Both read current across a shunt.

METER FRONT PANELS

- Both meters duplicate their terminals on the back. The 196 chooses between the two sets with a rear switch marked **INPUT**; the 34401A with a lit button on the front. If readings do not move, check that first.
- GPIB address on the 196:** press **PRGM**, **3**, **1**, set the number with the data entry keys, press **ENTER**. Factory 7, range 0 to 30.
- GPIB address on the 34401A:** press blue **Shift**, then **I/O**. Step to **HP-IB ADDR** with the arrow keys, set the number, press **Enter**. Factory 22, range 0 to 31.
- REMOTE** lights while the laptop drives a meter. **LOCAL** takes it back. The 34401A's current input has its own fuse, reachable from outside; a blown one reads zero current with nothing else wrong.

DO NOT. Never leave the cell connected and lit while 24 V is off. If 5 V is on and 24 V is off, the cell's current flows into the chip and destroys it. Connect the cell last and disconnect it first.